

1. OA (Day 1)
 - a. DSA
 - 20 mins
 - mathematical observation
 - b. Apti (MCQ) - 10
 - c. Systems (MCQ) - 15
 - OS
 - Networks
 - OOP
 - DBMS
2. Tech Interviews - 2 (Day 1 + 2)
 - a. introduce yourself, how long you've been programming
 - b. discuss projects on your CV
 - why did you choose a particular language for a particular project
 - I had an ML project, I was asked to explain neural networks
 - c. questions about the languages and tools you've used in your projects and mentioned in your CV
 - e.g. for python - what's the GIL, why it's needed, what are its implications. I talked a bit about cpython internals
 - rust - what's ownership, how is the idea useful
 - comparing different languages
 - examples of when you've used git, bash, powershell, etc
 - d. general questions about CS fundamentals
 - what happens under the hood between pressing power on on your computer and the desktop screen turning up
 - if you make a website what are the steps between a user typing in a url and then seeing your site in their browser
 - multiprocessing vs multithreading - pros and cons, when you'd use each, IPC
 - define polymorphism, encapsulation etc
 - e. DSA
 - leetcode medium level difficulty
 - actually write code on pen & paper
 - clarify problem statement, talk through your approach
3. HR Interview (Day 1 / 2)
 - a. what are your strengths and weaknesses
 - b. favourite aspect of swe
 - c. how would you rate yourself in this aspect
 - d. what do you plan to do after graduating
 - e. why did you apply to DE Shaw

General tips

- You won't know the answer to every question, that's fine

- You might be able to answer a few things from general knowledge, but don't make stuff up on the spot
- If you can't answer something on Day 1's interview, make sure to look it up before the next day
- Keep talking through the DSA portion even if you don't know the optimal solution immediately. If you have anything better than brute force it is worth mentioning, try pointing out which part could be optimized. Interviewers may give you pointers too
- If you know python, go, rust well, try to highlight those. If you don't, that is still completely fine, they care about your fundamentals far more than any specific language